

AI Career Coach Agent

An Intelligent Multi-Agent System for Personalized Career Guidance Using Google Gemini

Track

Concierge Agents

Problem Statement

Many aspiring professionals struggle to identify the skills required for modern AI and data careers. Information about learning resources, interview preparation, resume improvement, and required technical skills is often scattered across multiple platforms. As a result, learners spend significant time searching for information instead of following a clear and personalized learning path.

This project addresses that challenge by providing a single AI-powered career coaching system that guides users through every stage of their career preparation. From identifying skill gaps to generating learning roadmaps, reviewing resumes, and conducting mock interviews, the system delivers personalized recommendations tailored to each user's career goal.

Why AI Agents?

Career coaching involves several distinct tasks, each requiring different reasoning and expertise. Instead of relying on a single large prompt, this project adopts a modular multi-agent architecture where each agent is responsible for a specific task.

A central Orchestrator routes user requests to specialized agents, while a dedicated Security Agent validates and sanitizes user input before any AI processing occurs. This architecture improves modularity, maintainability, safety, and scalability, making it easier to extend the system with additional capabilities in the future.

Solution Overview

The AI Career Coach Agent provides an end-to-end workflow for career development.

The application allows users to:

Select a target career path.

Build and update their personal profile.

Analyze current skills against industry expectations.

Generate personalized learning roadmaps.

Receive ATS-focused resume feedback.

Practice technical interviews through a stateful mock interview system.

Save generated reports as Markdown files for future reference.

The application is implemented as a command-line interface (CLI) and uses Google Gemini to power intelligent responses while maintaining a secure and modular architecture.

Architecture

The AI Career Coach Agent follows a modular multi-agent architecture coordinated by a central Orchestrator. Every user request is first routed through the Security Agent, which sanitizes sensitive information using PII redaction and checks for prompt injection attempts before allowing further processing.

Once the request is verified as safe, the Orchestrator forwards it to the appropriate specialized agent based on the user's selected task. The system includes dedicated agents for Skill Gap Analysis, Learning Roadmap Generation, Resume Review, and Stateful Mock Interview Preparation. Each agent is responsible for a single domain, promoting clear separation of concerns, maintainability, scalability, and easier future expansion.

This architecture enables the application to provide personalized career guidance while ensuring that user interactions remain secure and well-organized.

Figure 1 illustrates the overall architecture and the interaction between the Orchestrator, Security Agent, and the specialized AI agents.

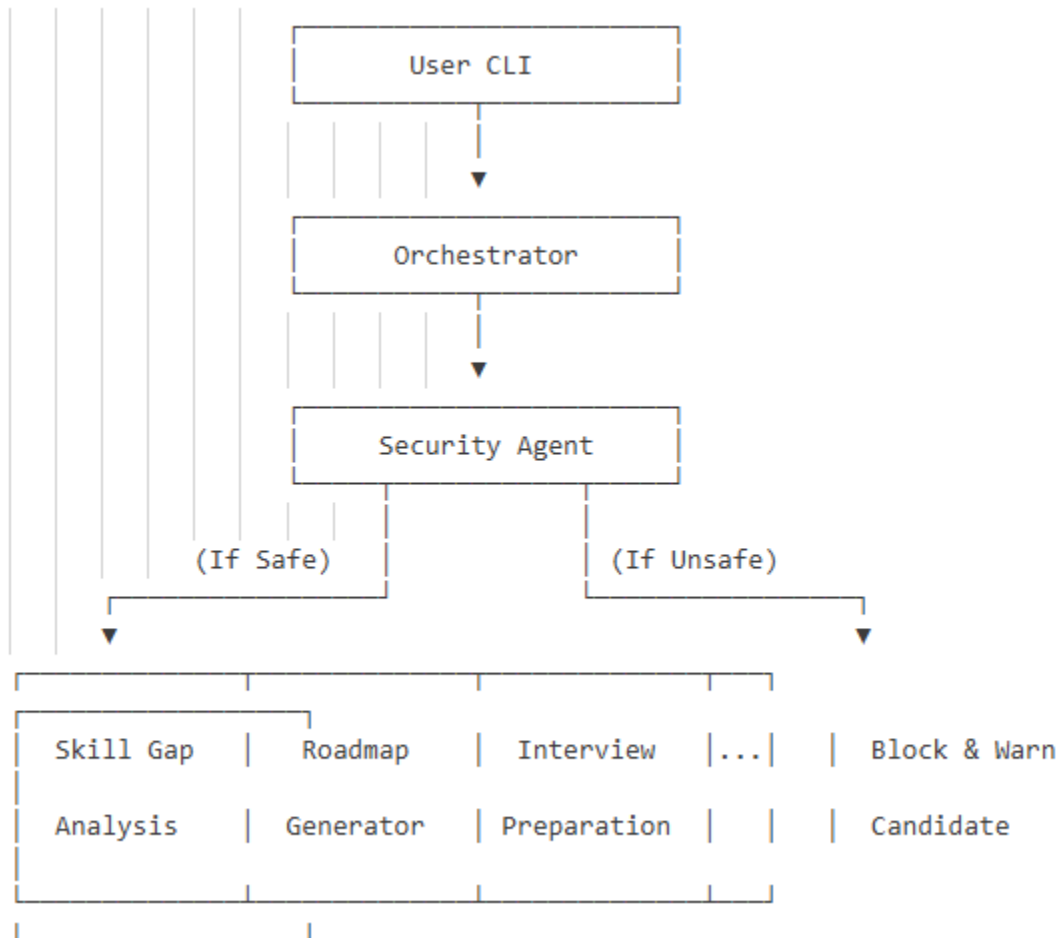


Figure 1. AI Career Coach Agent Architecture

System Workflow

The overall workflow of the application is illustrated below:

1. The user selects a target career path.
2. The Orchestrator receives the request.
3. The Security Agent validates and sanitizes the input.
4. The request is routed to the appropriate specialized agent.
5. Google Gemini generates the AI response.
6. The result is displayed in the CLI and optionally exported as a Markdown report.

Figure 2 shows the command-line interface (CLI) of the AI Career Coach Agent during execution. It demonstrates the interactive menu through which users can access the system's core functionalities, including profile management, skill gap analysis, roadmap generation, mock interviews, and resume review.

```
MAIN MENU - CURRENT GOAL: DATA ANALYST
=====
1. Set/Update Profile Details (Skills, Background)
2. Perform Skill Gap Analysis
3. Generate Learning Roadmap
4. Stateful Mock Interview Practice (3 Rounds)
5. Review Resume for ATS & Phrasing
6. Change Target Role / Career Goal
7. Exit

Select an option (1-7): █
```

Figure 2. AI Career Coach Agent CLI demonstrating the main interactive menu.

Key Features

- Multi-agent architecture coordinated by a central Orchestrator.
- Secure prompt handling with PII redaction and prompt injection detection.
- Personalized skill gap analysis for AI and data career paths.
- AI-generated learning roadmaps with milestone-based recommendations.
- ATS-focused resume review and improvement suggestions.
- Stateful mock interview sessions with personalized feedback.
- Session persistence for continued user progress.
- Markdown report generation for exported results.
- Modular and extensible project architecture.

Technologies Used

- Python 3.10+
- Google Gemini API

- Google GenAI SDK
- Antigravity IDE
- Visual Studio Code
- Git & GitHub
- python-dotenv
- unittest
- Regular Expressions (Regex)
- Markdown

AI Agent Concepts Demonstrated

This project demonstrates several key concepts covered during the Kaggle AI Agents course:

- Multi-Agent Architecture
- Central Orchestrator
- Security Guardrails
- Prompt Injection Detection
- Personally Identifiable Information (PII) Redaction
- Stateful Agent Workflow
- Google Gemini Integration
- Modular Agent Design

Challenges and Lessons Learned

One of the main challenges during development was designing a modular architecture that separates responsibilities across multiple specialized agents while maintaining smooth communication between them. Another challenge was implementing security measures to sanitize user input and detect prompt injection attempts before interacting with the language model.

This project also provided valuable experience in prompt engineering, agent orchestration, secure AI application development, Git/GitHub version control, and building a production-ready command-line application.

Future Improvements

Future versions of the project could include:

- A web-based interface using Streamlit.
- Integration with additional AI career paths.
- Voice-enabled mock interview sessions.
- PDF resume parsing and analysis.
- Database support for long-term user profiles.
- Analytics dashboard for tracking learning progress.
- Integration with MCP servers and external productivity tools.

Conclusion

The AI Career Coach Agent demonstrates how a secure multi-agent architecture can deliver personalized career guidance through specialized AI agents working together under a central orchestrator. By combining skill assessment, learning roadmap generation, resume optimization, and interview preparation into a single workflow, the project provides a practical example of applying modern AI agent design principles to solve a real-world problem.